AWI-ESM3 ↔ PISM orography and SMB coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-08-05

Abstract

Mission. Close the atmosphere half of the AWI-ESM3 ↔ PISM cycle in both directions: PISM's surface mass balance driven by the OIFS leg that just ran, and the ice-sheet surface elevation OIFS runs with following the ice sheet PISM produced. The ocean half was closed by the moving-cavity campaign; this is its companion.

Where we are. The SMB direction is closed. OIFS's own $P - E - R$ reaches PISM and PISM integrates on it, which retires the 130 ka taint of §1.1: the paleo file was driving the ice sheet 50% too wet. The ice sheet's discharge is wired back to the ocean as icebergs, built but not yet run. The orography direction has not started.

Timeline in §1, current state and open items in §2, dead claims in §3.2.

1 The campaign, in the order it happened

1.1 Taking stock: the atmosphere side is a stub (2026-08-03)

`couplings/oifs/just_copy_stuff.functions` copies `atmosphere_file_for_ice.nc` verbatim out of `a270122/lars_an` Confirmed in `eval5`: a 192×96 ECHAM Gaussian grid carrying `temp2`, `tsurf`, `aprt`, `orog`, `swd`, `cc`, `q2m`, `tau`, `emiss`, `uv`, `TOAswd`.

TAINTED: every PISM leg of the moving-cavity campaign, `eval5` and `fgdbg02` included, ran dEBM on a 130 ka paleo ECHAM atmosphere. No PISM surface mass balance number from that campaign is a statement about AWI-ESM3. Ocean-side results are not affected; they are forced from FESOM output directly.

Two further findings from the same read. `suorog.F90:151-250` already implements both returns: `ECE_CPL_ISMM` reads `usurf_oifs` and patches `ZZOROG` through `REESPE`/`SPEREE`, `ECE_LANDICE` reads `plit_oifs`. That code is upstream (vendor `a2e46d5`, Jorge Bernales) and our `suorog.F90` is byte-identical to it. But we run `ECE_LANDICE` only, and the `plit` we write is not PISM's ice mask: `make_plit_oifs.py` derives it from the regenerated ICMGG land–sea mask capped at 60°S. `eval5/couple/antar_pism2ece.nc` holds one variable, `plit_oifs`, `nx`=40320. No `usurf_oifs`. So no PISM geometry reaches the atmosphere at all.

1.2 Choosing the ISM-mapper over XIOS (2026-08-03)

Everything dEBM needs is already in the XIOS output, and the de-accumulation pattern exists in `field_def_lpjg_safe.xml.j2`. Checked against `eval5` year 1903: `ssrd` $= 6.845 \times 10^5$ and `tisr` $= 1.226 \times 10^6\,\mathrm{J\,m^{-2}}$ per one-hour step give 190 and 340 $\mathrm{W\,m^{-2}}$; `lsp+cp` $= 1.13 \times 10^{-4}$ m/step gives $3.14 \times 10^{-5}\,\mathrm{kg\,m^{-2}\,s^{-1}}$. All correct global means, so a jinja-gated `file_def` block would have delivered the forward direction with no new component.

Decided against it. XIOS solves the forward direction only: it cannot deliver `plit` into the live atmosphere, and cannot route basal melt and discharge into the ocean. Both need a component inside the OASIS exchange, and one mechanism is better than two. Staying on the EC-Earth component also keeps CISSEMBEL available.

Also decided: carry both SMB schemes with dEBM as default; SMB before orography; everything behind `general.smb_coupled`.

1.3 The ISM-mapper becomes an esm_tools component (2026-08-03)

`configs/components/ismm/ismm.yaml`, in the `develop-is` coupling, always built.

It is not a standalone repository: it lives inside `ecearth4` under `sources/ism-mapper`, so the component clones ecearth4 and builds that subtree. The clone is 37 MB, not the 663 MB the local `.git` suggests, because every `sources/*` model is a submodule that is not recursed into. The branch `cissembel-levante-intel` is required rather than preferred: it carries `5799a2a`, which fixes an `EBMfinalize` use-after-free that crashes any run with a non-CISSEMBEL region.

CISSEMBEL is a *link-time* dependency even on the direct path (`ism-mapper_run_mod.F90:7` has an unguarded `use modEBMworld`), so it is built from the same clone. At runtime `EBMworld` only fires for regions with `ISMRGNCIS(n)=.true.`, so it never executes for us.

The committed `Makefile` carries ECMWF HPC2020 paths. Everything it needs is a plain make variable and command-line macros beat file assignments, so the machine configuration is passed in `comp_command` rather than by patching a tracked file.

Verified. Built end to end from a fresh clone. The open risk was whether code written against OASIS 5.2 links against our 5.0-smhi; it does, with no source change. The binary carries `oasis_init_comp`, `oasis_def_var`, `oasis_enddef`, `ebmworld` and `ebminitialize`.

1.4 The PISM grid goes into ocp-tool (2026-08-03)

The ISM-mapper is an OASIS component, so the PISM domain must appear in `grids.nc`/`masks.nc`/`areas.nc`. This went into ocp-tool, not esm_tools: ocp-tool already sits between PISM and AWI-ESM3, it is where the weight generation lives, and `ocp_tool/grids/` already has the abstraction (a class with `cell_latitudes`, `cell_longitudes`, `cell_corners`, `cell_areas`, `cell_masks`). Putting it there deleted a planned step: ocp-tool writes `ismp` itself, so nothing appends fragments afterwards.

Centres come from the PISM file's own `lat`/`lon`; corners are inverse-projected from the cell corner (x, y); areas are spherical excess over two triangles.

Verified on `antarct_cr` 8 km, 761 × 761, three ways, because a wrong grid fails as silently-bad weights rather than a crash:

- Area total 36.03 against $36.28 \times 10^6\,\mathrm{km}^2$ from the plane area corrected by the polar stereographic scale factor, -0.70%.
- Cell areas $67.0\,\mathrm{km}^2$ at the pole to 53.8 at the outer corner against a nominal 64, the right sign and size for `lat_ts`$= -71$.
- All 579121 cells wind counterclockwise seen from outside the sphere, no sign flips. SCRIP requires this and the area check cannot detect it.

A full TCO95/CORE3 run puts `ismp` in all three files, bit-identical to an independently generated reference.

Found on the way: ocp-tool was appending the FESOM grid unconditionally. FESOM writes its own `feom` entry at runtime in domain-decomposition order; ocp-tool's is mesh order, and the two disagree. Now behind `ocean.write_oasis_grid`, default off. *This changes default output for every FESOM config, not only ours.*

1.5 Declaring the OASIS coupling (2026-08-03)

Field names on the ice side come from the ism-mapper itself (cplng_config_mod.F90): <prefix>_Precip_liquid, _Precip_solid, _Evaporation, _Runoff, _Soil4T, _SST inbound and <prefix>_plit out. On the OIFS side A_Evap_ISM, A_SST_ISM and A_Soil4T_ISM are what ecearth.F90 declares for the direct path; precip and runoff are reused from the ocean path.

Two constraints the parser caught, both of which would have been expensive later:

- A_Runoff is on atmr, not atma, so it needs its own exchange line and its own direction. Grouping it with precip fails the same-grid validation.
- include_models is consumed before general-level choose_ blocks resolve, so ismm must sit in the develop-is add_include_models. It moves together with smb_coupled: a launched ismm with no coupled fields deadlocks OASIS.

1.6 Generating the weights (2026-08-03)

Weights are not left to OASIS at model runtime. ocp_tool/oasis_weights.py precomputes them with pyOASIS taken from the coupled model's own OASIS build, ported from rdy2cpl: one MPI rank per link, each calling enddef() so SCRIP writes that link's rmp_*.nc. Geometry is read back from the OASIS description files, so the ismp entry of §1.4 is exactly what a link consumes. It needs compute nodes; the one-rank-per-link parallelism is what makes it tractable.

Tested on 3 nodes against ocp-tool's own TCO95/CORE3 output:

link	map	result
A096 → ismp	BILINEAR D	69 MB
R096 → ismp	GAUSWGT D	122 MB
ismp → A096	GAUSWGT LR	44 MB
ismp → A096	DISTWGT LR	34 MB
ismp → A096	BILINEAR	OASIS abort, rc=1
ismp → A096	CONSERV	OASIS abort, rc=1

Why the ice→atm conservative and bilinear maps die. 381 of the 579121 cells have a corner longitude spread above 180° along the polar seam, and the pole cell's four corners sit at $-135, 135, 45, -45$, encircling the pole outright. That breaks SCRIP's per-cell longitude line integral (CONSERV) and its enclosing-quadrilateral search (BILINEAR). Neighbour-based maps do not care. Same geometry as the reason the areas use spherical excess.

Consequence. plit reaches suorog as an interpolated value, not the area fraction ECE_LANDICE_THRESH is written to read. EC-Earth's bilinear would give the same, so we are no worse than the reference, but the threshold semantics are looser than ideal.

1.7 Running it: ten legs to a plausible number (2026-08-04)

Ten test legs, ismtest1 to ismtest10. The first six were plumbing, the last four were physics. Plumbing, briefly: the ism-mapper binary was stale because it had been built by hand rather than through esm_master and did not declare the namelist flags the generated namelist.ismm set; ismp was missing from the chunk-1 staged grids because chunk 1 stages from pism_cavity_ini/oasis/ and not from the fixed-mesh pool, so both pools now carry it; and LSendToRunoffMapper gated the ADD_FLD call but not UpdateISMForcing, which then looked for a field that had never been declared.

The mask wipe. ismtest6 and ismtest7 died the same way, at the clamp in ece_updclie_cpl.F90. updtim runs every timestep, the ISM coupling period is 86400 s and the OIFS timestep is 3600 s, so 23 steps out of 24 deliver nothing. On those steps the receive buffer still holds the `-HUGE` it was initialised to, and `CPLNG2_EXCHANGE` does not tell the caller that nothing arrived: it accepts `OASIS_Ok`, which means *call succeeded, no transfer*, in the same branch as the five statuses that do mean data arrived. An OASIS restart does not save you. It is read correctly at $t = 0$ and overwritten on the next step.

We fixed it on our side with `ECE_LANDICE_FROM_OASIS`, default `.false.`, so the OASIS-driven mask update is opt-in and `ECE_LANDICE` keeps reading `plit_oifs` at leg init. That is a workaround, not a fix. Any other `OASIS_IN` field whose coupling period exceeds the timestep has the same exposure. Issue drafted for the SMHI tracker.

Temperatures ten times too large. `ismtest8` delivered `antar_Soil4T` around 2600 K. The `gauswgt_avg` method had been copied from `gauswgt_c` and inherited its `CONSERV GLBPOS` postprocessing, which rescales the target field so the global integral matches the source. Source and target cover different areas, so the ratio is the whole error: A096 active area $359.41 \times 10^6\,\mathrm{km}^2$ over `ismp` active area $36.03 \times 10^6\,\mathrm{km}^2$ is 9.975. `CONSERV` belongs on a flux you are closing a budget with, between grids that cover the same domain. Neither holds here. Removed.

Too wet and too warm: the source grid was ocean-masked. `ismtest9` ran clean and produced `antar_smb_1900.nc` with no sentinels, which is the first end-to-end success. The numbers were still wrong, and the way they were wrong took a second look to see, because a first pass mistook the precipitation units. `A_Precip_liquid` and `A_Precip_solid` are divided by `RHOH2O` where they are set, so they are m/s. `A_Evap_ISM` is not, so it is $\mathrm{kg\,m^{-2}\,s^{-1}}$. Mixed units in one exchange line.

The comparison that matters is against OIFS's own gridded output from the same leg, over cells that are actually ice sheet, annual means in mm w.e./yr:

	P	E	R	T_{soil4}
OIFS `lsp+cp`, `e`, `ro` over its own ice cells	212	31	8.6	–
`ismtest9`, all ISM fields on `atma`	426	28.1	11.4	256.9 K
`ismtest10`, land fields on `atmr`	168	12.9	8.4	239.1 K

All three ISM-mapper rows are masked to PISM's own `thk`>10 m, 215030 cells, $13.8 \times 10^6\,\mathrm{km}^2$. `ismtest9` was 2.6 times too wet and 18 K too warm. Every ISM field had been declared on `atma`, the ocean-masked A096, where 95% of the cells south of 75°S are excluded; the GAUSWGT stencil then draws from what is left, which is the Southern Ocean. Ocean precipitation is heavier and the ocean is warmer, and both show up exactly that way.

`atmr` (R096) is the same 40320-point reduced Gaussian grid with the land mask instead, it is already declared in `oifs.grids`, `A_Runoff` already uses it, and the `R096` → `ismp` weights already exist. Moving `A_Evap_ISM` and `A_Soil4T_ISM` to it is a one-line config change. `A_Precip_liquid` and `A_Precip_solid` are not, because they are the same fields the ocean coupling uses, and the ocean needs them on `atma`.

`FIELD@grid`. EC-Earth solves this by naming a different grid per namcouple entry, which is what OASIS is built for. esm_tools cannot: `coupling_fields[field]["grid"]` holds one grid per field, full stop. The obvious workaround is to declare `A_Precip_*_ISM` aliases in OIFS, but that means editing code AWI-ESM3 shares with EC-Earth in order to work around a weakness in our own runscript engine, which is the wrong direction.

So the syntax gained a per-link override. `A_Precip_liquid@atmr` in a `coupling_target_fields` line uses `atmr` for that line and leaves the field declaration alone. It is 30 lines in `coupler.py`: one splitter, applied at each of the three places that parse a coupling string. Verified against the generated `namcouple`, where the ISM line now reads `R096 ismp` and the ocean line still reads `A096 feom`. esm_tools `b0555452f`.

`ismtest10`: the first SMB we believe. $P = 168$, $E = 12.9$, $R = 8.4$, so $\text{SMB} = P - E - R = 147$ mm w.e./yr over the PISM ice mask, 99.4% of the precipitation solid, mean soil layer 4 at 239 K. Verified in detail in §1.8.

1.8 Verifying the delivered field (2026-08-04)

Three questions. Does the remap preserve the field, is the field itself any good, and where does it sit against what is known about Antarctic precipitation in this class of model.

An orientation trap first. The PISM file's `y` coordinate variable runs opposite to its own data arrays. Forward-projecting the file's `lat`/`lon` at a given index lands at $-y[i]$, not $y[i]$. The file's `lat`/`lon` are the trustworthy pair: the northernmost ice cell they identify is at −62.93°, −60.56°, which is the tip of the Peninsula. Read straight off the `y` axis instead and the same cell comes out at −119°, in the Amundsen Sea, where no ice reaches that latitude. Sampling an OIFS field onto the PISM mask through the `y` axis therefore mirrors Antarctica through the pole, and because the ice sheet is roughly polar-centred the result still looks plausible. What settles it independently is the lapse rate. Correlating the delivered `antar_Soil4T` against PISM's own surface elevation gives −0.896 and −10.30 K/km as written, against −0.693 and −8.38 K/km mirrored. So the `ismp` grid ocp-tool built agrees with PISM's data arrays, and the delivered fields are in the right place.

The remap is faithful to 1%. Sampling every OIFS output cell whose centre falls inside PISM's `thk`>10 m mask, 7918 cells, and comparing against the same mask on the PISM side:

P [mm w.e./yr]	OIFS `lsp+cp`	delivered on `ismp`	difference
all ice	166.7	168.2	+0.9%
grounded margin, < 2500 m	245.3	247.1	+0.7%
grounded plateau, ≥ 2500 m	48.0	50.5	+5.2%

GAUSWGT is not conservative, so this is agreement rather than a guarantee, but at under 1% over the ice sheet the OASIS chain is not where any remaining error lives. E and R are not expected to match this closely and do not: `A_Evap_ISM` is tiles 3–9 only, weighted by tile fraction, while OIFS's `e` is the grid-box total including the ocean tiles.

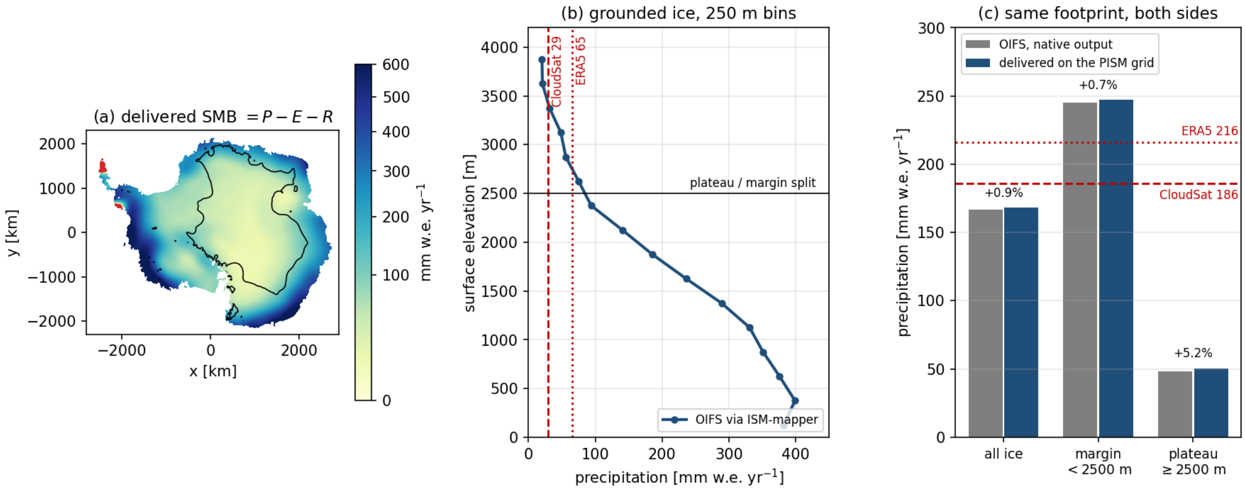

Figure 1: The first OIFS-driven surface mass balance, `ismtest10`, year 1900. (a) $P - E - R$ delivered on the PISM grid; the black line is the 2500 m contour and the red patches on the Peninsula are the only cells with negative SMB. (b) Precipitation against surface elevation over grounded ice in 250 m bins, falling monotonically from 391 mm w.e./yr below 500 m to 38 above 3000 m. The plateau values sit above CloudSat and below ERA5. (c) The same three footprints computed from OIFS's own output and from what arrived on the PISM grid. Agreement is within 1% over the ice sheet, so the remap is not where the error is. The continental mean, 167, is 10% *below* CloudSat, while ERA5 is 16% above it.

The field itself. Precipitation falls monotonically with elevation across seven 250 m bins, 391 mm w.e./yr below 500 m down to 38 above 3000 m, which is the right shape for an ice sheet fed by coastal cyclones. Integrated: 1725 Gt/yr over the 12.18×10^6 km^2 of grounded ice and 297 Gt/yr over the 1.59×10^6 km^2 of shelves, 2022 Gt/yr in total. Grounded SMB from RACMO is of order 2100 Gt/yr, so we are roughly 18% low, consistent with being 10% low on continental precipitation.

Against the literature. This is a known and well-documented bias family, so the numbers have a reference frame. Roussel et al. (2020, *The Cryosphere* 14, 2715) evaluated ERA5 against CloudSat: CloudSat gives 186 mm yr^{-1} continentally and ERA5 is about +30 mm yr^{-1} on top, and over the plateau the gap widens to 29 against 65. A 2025 ACP study found ERA5 and ten CMIP6 models all overestimate the frequency of supercooled-liquid-bearing clouds and the snowfall from them, with biases so similar across the ensemble that the authors attribute them to cloud microphysics rather than to the meteorology. Palerme et al. (2017) found the same overestimate across CMIP5.

Where we land:

P [mm w.e./yr]	CloudSat	ERA5 (IFS cy41r2)	ours (OIFS cy48r1, TCO95)
continental	186	216 (+16%)	167 (−10%)
plateau	29	65 (+124%)	50 (+72%)

So we are drier than ERA5 everywhere, below CloudSat continentally, and still carry a plateau wet bias, though a weaker one than ERA5's. The plateau comparison is the least secure part of this because CloudSat's near-surface blind zone is roughly the lowest kilometre, which is exactly where plateau clear-sky precipitation lives. The coastal numbers are on firmer ground.

The second mechanism in the literature is resolution, and it is the one that should worry us at TCO95. Genthon and Krinner made the margin-versus-plateau split the whole point: smoothed orography gives weaker lifting at the coastal escarpment, so less moisture is wrung out at the margin and more of it survives inland. Our dipole, low continentally but high on the plateau, is that signature. For an ice sheet this matters more than the continental mean suggests, because it moves accumulation off the fast-flowing outlet catchments and onto the dome, where it thickens ice that takes millennia to respond. A total that looks right can hide a pattern that is not.

1.9 The snow cap sends the same water to the ocean (2026-08-04)

`ece_fesom_set_ocean_fluxes.F90` caps the HTESSEL snowpack at 10 m water equivalent and sends everything above the cap to the ocean as `A_Calving`:

```
ZEXSNOW = MAX(SUM(PSP_SG(...)) + SUM(PSNSE1(...))*TSTEP - 10000.0, 0.)
PSURF%PSNSE1(...,1) = PSURF%PSNSE1(...,1) - ZEXSNOW/TSTEP
CPLNG2_FLD(CPLNG2_IDX('A_Calving'))%D(...) = (ZEXSNOW/TSTEP)/1000
```

This is the right closure for an atmosphere with no ice sheet underneath it. Snow falls on Antarctica, the pack would otherwise grow without bound, and returning the excess to the ocean mimics an ice sheet in balance calving what it accumulates.

The cap is saturated. Annual-mean snow depth over the PISM ice footprint is 9.952 m with a global maximum of 10.0004, and 98.7% of ice cells sit at or above 9.99 m. A saturated pack cannot store anything, so the shed rate equals the net input, which is the same $P - E - R$ we now hand to PISM. That puts `A_Calving` over Antarctica at roughly 2000 Gt/yr, against 2033 Gt/yr of delivered surface mass balance over the same mask.

So the water is counted twice the moment PISM is in the loop. It grows the ice sheet and it freshens the Southern Ocean, and only one of those can be right. There is no runoff-mapper diagnostic in the output, so this is inferred from the saturation condition rather than measured. Adding one is the cheapest way to close it.

The fix is to stop OIFS shedding over the PISM domain and let PISM's own discharge be what reaches FESOM instead. **That return path already exists.** `pism2esm` calls `pism2ocean` unconditionally, `iter_coup_interact_method_ice2oce` is already `BASALSHELF_WATER_ICEBERG_MODEL`, and `iterative_coupling_pism_ocean_prepare_ocean_icebergmodel_forcing` writes `latest_discharge.nc` from PISM's `tendency_of_ice_amount_due_to_discharge`. On the other side, fesom-2.7 already has `disch_file` pointing at exactly that file, behind `use_landice_water` and `use_icesheet_coupling`.

The whole chain is switched off by one flag. `iceberg_coupling: 0` in the runscript becomes `PISM_TO_OCEAN=0` and `fesom_use_iceberg=0`, and `pism2ocean` returns in 0.06 s with *NOT generating ice forcing for ocean model*. So closing the double count is a configuration change and a test, not a piece of development.

Note what the selected method deliberately leaves out: the basal-shelf line is commented out, so only discharge returns, not basal melt. That is right for us. FESOM computes the cavity melt itself and PISM consumes it through `-ocean given`, so sending PISM's basal melt back would double count in the other direction.

What is genuinely untested is whether it runs. `use_icebergs` is `.false.` in every FESOM namelist we ship, and the helper that would tell the iceberg module how many bergs to seed, `get_ib_num_after_ice_sheet_coupling`, is commented out.

This does not taint the moving-cavity results. Without PISM returning mass, the shedding was the correct closure and the ocean saw a balanced ice sheet, which is what those runs assumed.

1.10 PISM eats what OIFS gives (2026-08-05)

`just_copy_stuff.functions` copied `atmosphere_file_for_ice.nc` out of another user's 130 ka experiment (§1.1). It is gone. `smb2ice` hands over the ISM-mapper's own output instead, and falls back to the old copy only when `smb_coupled` is off.

On the ice side, `ablation_method: DIRECT` builds the surface-given file straight from the atmosphere's water budget, climatic_mass_balance $= (P_{\text{liq}} + P_{\text{sol}} - R)\rho_w - E$, with `ice_surface_temp` from soil layer 4. No melt model: over Antarctica HTESSEL's budget already *is* the surface mass balance. `regrid_method: NONE`, because the ISM-mapper delivers on the ice sheet's own grid.

Two things surfaced while building it. The pool mask for `antarct_cr` is a symlink to a 192×96 ECHAM grid, so it cannot mask a 761×761 field at all; the mask now comes from `thk` and `topg` in PISM's own state. And without any mask the coastal values the GAUSWGT stencil smears over the Southern Ocean would nucleate ice offshore.

What changed for PISM. Against `ismtest10`, the last run on the paleo file, with the same PISM configuration:

	dEBM on the 130 ka file	DIRECT on OIFS's $P - E - R$
SMB tendency [Gt/yr]	$3103 \rightarrow 3106$	$2010 \rightarrow 1984$
discharge [Gt/yr]	$-35345 \rightarrow -9578$	$-28628 \rightarrow -8342$
basal mass flux [Gt/yr]	$-1428 \rightarrow -817$	$-1438 \rightarrow -805$

The paleo file drove PISM at 3103 Gt/yr against an observed Antarctic 2100–2400. So the taint was worth about +50%, and the replacement is inside the observed range. Basal mass flux is unchanged to 1%, which is the control: it comes from FESOM's cavity melt, which this did not touch.

Not a statement about the ice sheet. Both runs shed 8000–9600 Gt/yr and carry a discharge 3–4 times an in-balance Antarctica even at year 10. That is the inherited spinup relaxing, not the coupling. §1.13 deals with it.

1.11 The mesh regeneration stopped writing the ocean grid (2026-08-05)

`ismtest11`'s second cycle died on every one of 1792 ranks:

```
MCT::m_SparseMatrixPlus:: FATAL--length of vector x different from column count
of sMat. Length of x = 208432   Number of columns in sMat = 208810
```

208432 is the new mesh, 208810 the old. The chunk-2 namcouple correctly declared the new size while all three staged `rmp_*feom*.nc` still carried the old one.

The cause was ours, from two days earlier. Gating ocp-tool's FESOM export behind `ocean.write_oasis_grid` is right for the initial pool, where FESOM writes `feom` itself at runtime. It is wrong for the regeneration, which needs the grid present precisely so the new weights can be built against it. `permute_feom_to_runtime` then found nothing to permute and refused to continue, correctly.

A second bug, not ours, and worse. `couple_in` did exit 1. The generated run script backgrounds it and lets `observe_couple_in` submit the next leg regardless of its status:

```
ice2fesom > log 2>&1 &
process=$!
esm_runscripts ... -t observe_couple_in -p ${process} ...
wait
```

So a cleanly reported failure still cost a 32-node MCT abort. Left alone, since it is shared `esm_runscripts` behaviour for every setup. Mitigated instead by a post-condition: after regeneration the new weights are checked against `nod2d.out` and a mismatch is named with both sizes rather than reaching MCT.

1.12 Which way the ice sheet's water reaches the ocean (2026-08-05)

Two routes exist. The runoff mapper can take the ISM's basal melt and discharge directly, summing them into runoff and calving. Or the discharge can become icebergs.

The first was built and then thrown away. It worked, and the arithmetic checked out: converting PISM's gridded $\mathtt{kg}\,\mathrm{m}^{-2}\,\mathtt{yr}^{-1}$ to m/s and integrating gave -11222.3 Gt/yr of discharge against -11222.3 in PISM's own `ts` file. But it ran PISM's discharge through the feedback file, the ISM-mapper and a new OASIS link to arrive where the runoff mapper could have been handed it, and it is not the route we want physically. Icebergs drift and melt over months across a wide area; a runoff-mapper flux is deposited at the calving front immediately.

Kept from that work: the runoff mapper upgrade. Ours was an ancestor of EC-Earth4's, four coupling fields and a three-entry namelist. The new one brings `IgnoreCalvingBasins`, which is needed whichever route the water takes: the atmosphere's snow-cap calving over Antarctica has to stop, or it is counted twice (§1.9). Basin 66 is Antarctica exactly, 13383 land cells from 89.5°S to 64.2°S, none north of 60°S, and 1537 ocean calving points. Our one local change, dropping the calving enthalpy conversion, was already upstream.

The iceberg route, and why not the plugin. `fesom_icb_pism` registers `esm_tools.plugins` entry points, and `esm_plugin_manager` imports every registered plugin at startup. Installing it pulled `pyfesom2` into every `esm_runscripts` call, icebergs on or off, and on this machine that stack does not import: `libproj` wants a `libstdc++` newer than `/lib64` has, and `cmocean` calls `matplotlib.cm.register_cmap`, gone since matplotlib 3.9. An unrelated config check died because of an iceberg plugin.

So the two halves are split by weight, the way ocp-tool is already handled. The generator needs `pyfesom2` and runs as a subprocess under the ocp-tool python, from `couple_in`, right after `latest_discharge.nc` is written. Counting the bergs into `icebergs%ib_num` needs nothing and is a core prepcompute step with no entry point.

No berg set is seeded. `icb_felem` indexes the mesh, and the mesh changes every chunk, so the set is rebuilt per leg from that leg's discharge and mesh. Run 1 starts empty, because PISM has not run yet.

1.13 Starting PISM from a relaxed state (2026-08-05)

The discharge is the iceberg forcing now, so the spinup transient stopped being a background annoyance. Over a270234's `repro_movcav45`, 570 years of coupled PISM on the same 8 km `antarct_cr` grid:

chunk	discharge	basal	SMB	Δmass
year 0	−14253	−1009	3099	−9763
year 10	−9729	−784	3112	−7338
year 530	−3966	−713	3152	−1448
year 560	−4141	−685	3157	−1666

Discharge falls by 3.4 and the mass loss by nearly 6. The year-560 restart is now the start state. Still above a balanced ~2000 Gt/yr, but in a range where the ocean response is worth reading; the old spinup would have delivered roughly five times that.

1.14 Bringing up icebergs: six defects (2026-08-05)

The freshwater route is icebergs, not the runoff mapper's own ISM inputs (§1.12). Getting a leg to start took seven submissions. Five of the six things in the way were already broken; one was ours. Listed because the pattern says something about how much of this path had been exercised before.

1. **The plugin poisons every run.** `fesom_icb_pism` registers `esm_tools.plugins` entry points, and `esm_plugin_manager` imports every registered plugin at startup. Installing it pulls `pyfesom2` into every `esm_runscripts` call, icebergs on or off. On this machine that stack does not import: `libproj` needs `GLIBCXX_3.4.26` and `/lib64` stops at 3.4.25, and `cmocean` calls `matplotlib.cm.register_cmap`, removed in matplotlib 3.9. An unrelated config check died because of an iceberg plugin. Proved by `LD_PRELOAD` both ways: the old `libstdc++` first reproduces it, the conda one first fixes it.

2. **`with_icb` cannot be set from a `choose_` block.** fesom's `choose_with_icb` resolves first, so the whole iceberg block stays off while every flag reads true. It goes in the driver runscript beside `with_recom`.

3. `bleach 5.0.0` in the shared spack python has a malformed requirement, `tinycss2 (>=1.1.0<1.2)`, missing a comma. It breaks `pkg_resources` whenever plugins are scanned, which is exactly when `with_icb` is on. Upstream's 5.0.1 changelog entry is *Add missing comma to tinycss2 require*.

4. **`prepcompute_recipe` under `choose_with_icb` is never read.** The live recipe is `general.prepcompute_recipe`; the fesom one sits in the finished config unused, 17 steps against 19. So `prep_icebergs` and `apply_iceberg_calving_to_namelists` have *never run for anyone*, and the plugin's only effect to date has been the import of item 1.

5. **`icb_calving_day.dat` is missing everywhere.** FESOM opens it `status='old'` and reads `ib_num` values, in both `init_icebergs` and `init_icebergs_with_icesheet`. It is in neither `icb_inputs` nor the staging lists, and `_icb_generator` does not write it: that writes longitude, latitude, length, height, scaling and felem. Found the format in a working ini directory of a270124's, one day-of-year per berg.

6. **Ours.** A config file staged into `work/` takes the basename of its *source*, so a template named `namelist.runoffmapper.ismm` lands under that name and the runoff mapper dies on a missing `namelist.runoffmapper`. Fixed with a versioned directory rather than a rename, since `config_in_work` set from `choose_version` does not merge either.

What we built instead of using the plugin. The two halves are split by weight, the way ocp-tool is already handled. `make_icebergs.py` needs `pyfesom2` and runs as a subprocess under the ocp-tool python from `couple_in`, right after `latest_discharge.nc` is written. `count_icebergs_into_namelist` counts them into `icebergs%ib_num` and is a core prepcompute step with no entry point, so nothing heavy is imported unless bergs are actually being made.

That counter cost two of the seven submissions on its own. It has to write to `namelist_changes` and run *before* `modify_namelists`, which loads, modifies and writes in one step. Editing the loaded object afterwards, which is what the upstream plugin does, changes nothing on disk. The first version was right and sat in the dead recipe of item 4, so it logged the correct `ib_num = 0` while the file on disk kept the template's 1.

Three more, found by running it. The count reached nine.

7. **`icb_calving_day.dat` is missing everywhere** (already listed as 5 above).

8. **Zero bergs kills FESOM.** `init_buoy_output` defines `number_tracer` with extent `ib_num` and then `time` as `NF_UNLIMITED`. In classic netCDF a zero-size dimension *is* unlimited and only one is allowed, so the run dies at step 1 with `NC_UNLIMITED size already in use`. `ib_num` is never tested against zero anywhere in the iceberg code; the stepping itself is fine with it. Fixed by guarding the five output call sites, FESOM/fesom2#962. Verified: before, dies at step 1; after, reaches step 3649 and integrates normally.

9. **The generator needs cell corners nothing writes.** `_get_coords` reads `lon_bnds`/`lat_bnds` to place bergs inside a discharge cell. The PISM griddes in the pool carries centres only, so `cdo -setgrid` cannot add them and `latest_discharge.nc` comes out without: the real pipeline would have failed the same way. `add_cell_bounds.py` supplies them from ocp-tool's own `PISM.cell_corners`.

The worst one was silent. `count_icebergs_into_namelist` read the couple directory, which is right from run 2, when the generator writes there. On run 1 the bergs come from the seed directory instead, so a staged set of 63015 came out as `ib_num = 0` and FESOM would have ignored every one of them. No crash, no warning, just a run with no icebergs that looked fine. The zero-berg guard of item 8 is exactly what would have let that pass unnoticed.

The generator works. 63015 bergs from the discharge of the same movcav45 state the PISM restart comes from (§1.13). All seven files consistent, `icb_felem` inside the mesh's 405888 elements, latitudes −76.4 to −60.5 along the Antarctic coast, total mass of order 2800 Gt against that state's 4141 Gt/yr. Banked as the chunk-1 set, valid for that mesh only. `scaling` is 1 throughout, so these are 63015 individual bergs rather than a smaller set standing in for more; the awiesm-2.1 reference compressed the count with `scaling_factor [100, 50, 10, 1, 1, 1]`, worth revisiting if advection is slow.

How many bergs. A PISM discharge of 4141 Gt/yr produces 63015 bergs at `scaling_factor` 1, and 72% of them are under 0.1 km^2. Measured against an otherwise identical leg with `ib_num=0`: 0.0948 against 0.0754 s per FESOM step, so the uncompressed set costs +26%, about 33 minutes against 26 for the year.

That is affordable, and the first instinct was to keep the sampling and pay it. The code says otherwise. `icb_thermo.F90` scales the flux to the ocean by `scaling(ib)`:

```
hfb_flux_ib(ib) = H_b * length_ib*length_ib*scaling(ib)
dh_b = M_b*dt*REAL(steps_per_ib_step)   !*scaling(ib)
```

but deliberately not the decay. A model berg melts at the rate of a berg of its chunk's *mean* size, and melt is non-linear in size, so the factors change the answer. They are a physics parameter, not a cost knob, and an unrun value is an unchecked one. So the awiesm-2.1 values `[100, 50, 10, 1, 1, 1]`: 1227 model bergs standing for 65219 real ones, mass conserved exactly at 2756 Gt.

The CAV-ICB reference goes further, `[250, 100, 30, 2, 1, 1]` for 485 bergs, on a finer mesh where advection bites harder. The +26% measurement is recorded here so that sampling more, if it is ever wanted, is an experiment against a baseline rather than a guess.

Finn Heuer's CAV-ICB plugin solves the neighbouring problem: it derives the calving mass as a residual of FESOM's own cavity melt minus accumulated Antarctic snow, which is what you need when there is *no* ice sheet model. With PISM the discharge is known, so its core is not for us. Three of its downstream details are better than ours and worth taking: calving days with an austral summer intensification for the small classes rather than a flat spread, `exclude_occupied_elements`, and a considered `ibareamax` of 100 against the inherited default of 400.

Still not verified. No leg has generated bergs *in run*. Chunk 2 is the first, and it is also the leg that exercises the `feom` fix of §1.11.

2 State of the coupling (2026-08-05)

2.1 Proven

- ISM-mapper builds from a clean clone and links against our OASIS 5.0-smhi.

- `ismp` grid written by ocp-tool into all three OASIS files, geometry verified three ways (§1.4).
- All three coupling links generate on compute (§1.6).
- The esm_tools config assembles: `ismm` in `process_ordering`, `rstism.nc` a recognised OASIS restart, four exchange lines resolving with the prefix substituted.
- A six-component OASIS `enddef` succeeds with ISMMAP alongside OIFS, FESOM, XIOS, LPJ-GUESS and the runoff mapper.
- A full year runs coupled and writes `antar_smb_1900.nc` on the PISM grid, monthly, with no sentinels and no NaNs (§1.7).
- Daily coupling with `LOCTRANS AVERAGE`, so 365 exchanges a year rather than 8760.
- PISM integrates on it. `just_copy` is gone, and the paleo file it served was driving PISM at 3103 Gt/yr against an observed 2100–2400 (§1.10).
- The delivered SMB is physical: 147 mm w.e./yr over the ice mask, 2022 Gt/yr in total, 99.4% solid precipitation, monotonic in elevation across seven bins (§1.8).
- The remap preserves precipitation to within 1% over the ice sheet, checked against OIFS's own output on an identical footprint (Fig. 1c).
- The delivered fields are correctly oriented, confirmed independently by a −10.3 K/km lapse rate against PISM's own surface elevation.

2.2 Open

- **Plateau wet bias, margin dry bias.** +72% against CloudSat above 2500 m while 10% low continentally (§1.8). Expected at TCO95 and shared with ERA5, but it redistributes accumulation from the outlet catchments to the dome, which is the pattern PISM is most sensitive to. Nothing to fix today; something to state whenever a run is interpreted.
- **Nothing has run with icebergs on.** The generator imports and the call site is wired, but no leg has invoked it on real discharge. Chunk 1 makes none by design, so chunk 2 is the first real test, and it is the same leg that exercises the `feom` fix of §1.11.
- **A failed `couple_in` still cannot stop the chain** (§1.11). Left alone deliberately; it is shared `esm_runscripts` behaviour.
- **HTESSEL glacier-tile refreezing** is unchecked. It decides whether the shelf-margin SMB is meaningful before PISM ever sees it.
- `A_Calving` **double-count, about 2000 Gt/yr** (§1.9). Confirmed: the snowpack is saturated at the 10 m w.e. cap over 98.7% of the ice sheet, so everything that accumulates is shed to the ocean, and the same water now also grows PISM. Must land together with the freshwater return, never before it.
- `orog` **for the downscaling reference.** `field_def.xml:176` has a `z` but it is in the `3D_ml` group, so model-level geopotential. Source it from the ICMGG with eccodes for param 129, as `make_plit_oifs.py` does for `lsm`.
- `usurf_oifs` **writer**, replacing the ICMGG-derived `plit` with PISM's own mask, and turning `ECE_ISM_OROG` on.
- `PLIT_antar` **is received on** `atma`, because `ismp` → `R096` weights were never generated. Harmless while `ECE_LANDICE_FROM_OASIS` is off, wrong the moment it is turned on for the orography half.

- **Freshwater return.** PISM basal melt and discharge do not reach FESOM. Not needed for SMB or orography, but it is the remaining leg of the mass budget.

- **Field set is welded to scheme in ismm.** The direct path supplies 6 fields, dEBM needs 7 and shares only 2 of them. Splitting the two would unblock dEBM for Greenland and for warm scenarios where a positive-degree-day treatment stops being adequate.

- **atlas 0.31.1 → 0.39.0** came in with the vendor merge and is unbuilt.

3 Reference

3.1 Secondary fixes

what	where	commit
`mod_domain.F08` listed in `MOD_SRCO` but absent, CISSEMBEL unbuildable	ecearth4	`41e0d8e`
Runoff / runoff-mapper exchanges made optional (`-huge` trap)	ecearth4	`f9d0349`
`ECE_ISMM_SET_STATE` call had no `intfb` include	oifs-48r1	`38c3ab0`
`debm` missing from `add_other_components`, esm_master aborted	esm_tools	`55a7935`
`LSendToRunoffMapper` gated `ADD_FLD` but not `UpdateISMForcing`	ecearth4	`94511fb`
OASIS-driven land-ice mask update made opt-in	oifs-48r1	`d23116d`
`gauswgt_avg` inherited `CONSERV GLBPOS` from `gauswgt_c`	esm_tools	`c47fea07e`
Per-link source grid override, `FIELD@grid`	esm_tools	`b0555452f`

The `-huge` trap is worth stating once in full: receive buffers are initialised to `-huge`, `AccumulateECEForcingFields` adds every `OASIS_IN` field with no check that it is coupled, and an uncoupled field is never written. So a declared-but-unpartnered input accumulates `-huge` for a whole leg. This is the same failure class as the OIFS-side `-HUGE` first-send bug, arriving from the other end.

3.2 Falsified claims (kept, struck through)

- ~~Adding a PISM grid means teaching the per-leg regeneration to emit SCRIP weights to and from TCO95 on every cycle. This is the expensive part and the main schedule risk.~~ Overstated: the weight machinery is general, so PISM costs a few `Link` entries.

- ~~OASIS computes the weights itself at runtime, and the PISM grid does not move, so there is no per-cycle work.~~ Wrong twice. Weights are precomputed via pyOASIS (§1.6), and the links do regenerate per mesh-change leg because the OIFS mask moves where PISM is.

- ~~CONSERV is the right map for plit, since ECE_LANDICE_THRESH reads an area fraction.~~ It aborts on a pole-centred grid (§1.6). GAUSWGT is what works.

- ~~The ice→atm link is `BILINEAR D`.~~ The SCRIPR letter is the *source* grid type and `ismp` is logically rectangular, so `LR`; and `BILINEAR` fails here regardless.

- ~~`plit-active=0` in the log is the symptom of the mask being wiped.~~ That line appears 363 times a leg in `eval5`, which runs fine. It is a normal diagnostic and says nothing about the crash. The evidence for the wipe is the clamp itself and the delivery arithmetic, not this line.

- ~~All ISM source fields belong on `atma`.~~ `atma` is ocean-masked. Everything except SST belongs on `atmr` (§1.7).

- ~~The delivered P is 168 against OIFS's own 212, so the remap may be losing precipitation.~~ The 212 came from a snow-depth proxy mask on the remapped output, which is coastal-biased. On an identical footprint the two agree to 0.9% (§1.8).

- ~~OIFS sits at ERA5's continental Antarctic precipitation of about 216 mm/yr.~~ Same proxy-mask error. It is 167, which is 10% *below* CloudSat.

3.3 Method notes

- `esm_runscripts -c` assembles the full runtime config and catches grid-consistency and component-loading errors. Both config bugs above were found this way, not by reading.
- `esm_master -check <target>` renders the build without executing it. It also revealed the pre-existing `debm` section-header failure.
- Weight generation must run from a shared filesystem. A first attempt with the working directory on `/tmp` died in 14 s: `/tmp` is node-local on Levante, so the compute node could not write the job output.
- **A `choose_` block cannot override what another `choose_` block sets.** This cost three separate rounds: `with_icb` set from `choose_general.ism_freshwater_return` left the whole iceberg block off while every flag read true; `required_plugins` and `prepcompute_recipe` set from the setup lost to fesom's `choose_with_icb`; `config_in_work` set from `choose_version` never merged. Each time the fix was to move the value to where it resolves first, not to try to outrank it. Literals in the driver runscript win, and a versioned directory beats renaming a file.
- The name a config file gets in `work/` is the *basename of its source*, not the key it is filed under. A template named `namelist.runoffmapper.ismm` is staged under that name and the model dies on a missing `namelist.runoffmapper`.
- Grid winding is invisible to an area check, which takes an absolute value. Test it separately by dotting the corner-polygon normal with the outward radial vector.
- Do not guess the units of an OASIS field. Read the line that sets it. `A_Precip_*` are divided by `RHOH2O` and are m/s, `A_Evap_ISM` is not and is $\mathrm{kg\,m^{-2}\,s^{-1}}$, and they travel in the same exchange line. Guessing wrong turned a factor of 2.6 into an apparent factor of 350 and pointed at the wrong bug.
- Compare a remapped field against the source model's own output over the *same* mask, not over a latitude band and not over a proxy for the mask. Both shortcuts produced a wrong answer here, in opposite directions.
- A polar grid can be mirrored through the pole and still look right, because the ice sheet is roughly polar-centred and latitude is unchanged by the mirror. Longitude is not, so check a known asymmetric landmark. The Peninsula tip at $-62.9°$, $-60.6°$ settled it here. An independent physical check is better still: temperature against elevation gave -10.3 versus -8.4 K/km for the two orientations.

3.4 Assets

what	where
Reference document	`esm_tools/ece4-pism-coupling.md`
Verification figure script	`scripts/plotting/plot_smb_ismtest10.py`
First believed SMB	`ismtest10/outdata/ismm/antar_smb_1900.nc`
`cplng2 -HUGE` issue draft	`/work/ab0246/a270092/tmp/oifs_issue_draft.md`
PISM OASIS grid (reference copy)	`/work/ab0246/a270092/input/pism/oasis/`
Weight-generation test output	`/work/ab0246/a270092/tmp/ismp_wgen{,2}/`
ocp-tool ice-enabled config	`configs/TCO95_CORE3_ICE.yaml`

repository	branch	state
esm_tools	awiesm3-is-orography-and-smb-coupling	b0555452f pushed
ocp-tool	feat/pism-oasis-grid	PR #57, draft
oifs-48r1	movcav-landice+co2-concdriven	460015f pushed
ecearth4	cissembel-levante-intel	pushed